

PRISM: Physics-Informed Reconstruction via Implicit Surfaces and Materials

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Abstract

Recovering 3D geometry, surface material, and lighting from sparse 2D images is fundamentally ill-posed: existing implicit-surface approaches either decouple geometry from appearance or adopt Lambertian shading that fails on specular and metallic objects. I present PRISM (Physics-Informed Reconstruction via Implicit Surfaces and Materials), a feed-forward multi-view model that jointly infers a neural Signed Distance Function (SDF), Cook-Torrance GGX BRDF parameters, and a point-light model from one or more RGB views. PRISM couples geometry and appearance through NeuS-style differentiable volume rendering: the photometric loss backpropagates through the implicit surface into the SDF, BRDF head, and light head simultaneously, while per-frame depth, normal, silhouette, visual-hull, and SDF sign-band constraints anchor the implicit field. Trained and evaluated on the OmniObject3D dataset of real-world scanned objects, the full model achieves a mean Chamfer distance of 0.209 and a mean foreground PSNR of 14.55 dB across 19 test objects spanning 6 categories. Loss-group ablations show that photometric supervision is the largest single contributor to appearance quality (PSNR drops 4.5 dB when removed), depth supervision is second (1.1 dB), and normal and eikonal terms contribute under 0.25 dB each.

1. Introduction

Disentangling 3D geometry, surface materials, and scene illumination from a small set of 2D images is fundamentally ill-posed. A camera collapses the 3D world into a 2D grid, merging shape, material, and lighting into a single per-pixel radiance value with no record of which physical factors contributed which fraction. Many learned systems sidestep this by predicting geometry alone, with no compatible model of reflectance or lighting [7, 5, 4]; their reconstructions cannot be relit or composited into new scenes. Methods that do model appearance — NeRF-style radiance fields — instead conflate shape with view-dependent emitted color [9, 13], so geometry is entangled with the lighting under which it was observed and cannot be decomposed into reusable physical components.

Resolving these ambiguities through inverse rendering — recovering shape, material, and illumination from observations — matters for downstream use. Autonomous robotics benefits from physically consistent reconstructions for compact, on-device 3D understanding; AR/VR and content creation require geometry and material representations that can be inserted into novel lighting without artifacts. Bridging raw pixels and physically principled 3D properties is a necessary step

toward vision systems that treat the world as a physical environment rather than a flat canvas.

Neural Radiance Fields (NeRF) [9] showed that coordinate-to-radiance MLPs trained with differentiable volume rendering produce high-quality novel-view synthesis without explicit geometry, and PixelNeRF [13] generalized this feed-forward by conditioning on convolutional image features. But radiance fields conflate shape with view-dependent emitted color, and follow-ups confirm the internal decomposition is not directly usable as a physical material model: Ref-NeRF [11] reparametrizes outgoing radiance into reflected-view coordinates, and NeRFactor [6] distills a pre-trained NeRF into albedo, normals, and visibility for relighting.

Concurrent implicit-surface work has pursued cleaner geometry through signed distance functions. IDR [4] differentiates through sphere tracing via the implicit function theorem; NeuS [8] replaces hard sphere tracing with logistic-density volume rendering for smooth gradient flow through silhouettes and thin structures; VolSDF [14] independently proposed a related density-from-SDF formulation. Together these establish that volume-rendered SDFs combine the geometric clarity of

implicit surfaces with the gradient stability of NeRF-style training.

Despite this progress, principled physical material models and feed-forward implicit surfaces have remained largely separate. Existing physics-based systems either operate per-scene [12, 6], require a pre-trained NeRF as intermediate representation [6], or only target view-dependent radiance reparameterization without explicit material parameters [11]. The ICCV 2023 OmniObject3D challenge winner [3] is feed-forward but stacks depth supervision atop a PixelNeRF-style radiance field, achieving strong geometry without principled material decomposition. The gap motivating this work is a feed-forward system that jointly recovers an explicit signed distance surface, a physically based BRDF, and a lighting model in a single forward pass.

I propose PRISM (Physics-Informed Reconstruction via Implicit Surfaces and Materials) to fill this gap, training and evaluating exclusively on OmniObject3D [1] (~6,000 real-world scanned objects across 190 categories with Blender-rendered RGB, depth, and normal maps and ground-truth textured meshes). The principal contributions are: (i) a feed-forward system for joint geometry, Cook-Torrance GGX material, and point-light estimation via differentiable NeuS rendering; (ii) a multi-loss objective coupling photometric, geometric, silhouette, SDF sign, visual-hull, and eikonal terms; (iii) a PixelNeRF-style local feature projection that conditions the SDF MLP on spatially indexed input-view features; and (iv) a loss-group ablation study quantifying the contribution of photometric, depth, normal, and eikonal supervision. Code is available at <https://github.com/osbo/prism>.

2. Related Work

2.1. Implicit Neural Surface Representations

Implicit neural representations have largely displaced explicit voxel and mesh approaches. Occupancy Networks [7] represent shape as the level set of an MLP, and DeepSDF [5] extends this to signed distance functions for cleaner gradient information. Both established that latent-conditioned MLPs generalize across categories given sufficient 3D training data.

Connecting an implicit surface to image observations through a differentiable forward model is the core challenge for learned inverse rendering. IDR [4] differentiates through sphere tracing using the implicit function theorem, enabling photometric supervision of the SDF. NeuS [8] replaces hard sphere tracing with logistic-density volume rendering, accumulating color and depth from a continuous distribution of samples concentrated at the SDF zero crossing — important early in training when the SDF is coarsely initialized. VolSDF [14] independently proposed a related density-from-SDF formulation using a Laplace CDF.

2.2. Feed-Forward Generalization and Multi-View Conditioning

NeRF [9] requires per-scene optimization from many calibrated views. PixelNeRF [13] makes the formulation feed-forward by conditioning the radiance field on convolutional image features projected from image space to 3D query points using known camera geometry, providing local appearance context that improves fine surface detail.

The ICCV 2023 OmniObject3D challenge winner [3] is the most directly comparable prior work on the dataset used here. The approach stacks explicit depth supervision on top of a PixelNeRF-style radiance field, achieving strong geometric reconstruction by grounding the implicit field in observed depth, but does not target principled material decomposition.

2.3. Inverse Rendering and Physics-Based Material Models

Inverse rendering jointly recovers geometry, reflectance, and illumination from images — three unknowns coupled in each pixel observation, harder than geometry-only reconstruction. Sengupta et al. [10] jointly estimate albedo, normals, and indoor lighting from a single image under Lambertian shading. PhySG [12] uses mixtures of spherical Gaussians for both BRDFs and environment lighting, recovering high-quality materials on glossy objects per-scene. NeRFactor [6] decomposes a pre-trained NeRF into albedo, normal, and visibility for relighting, again per-scene.

The Cook-Torrance GGX microfacet model [2] captures specular reflection, geometric shadowing-masking, and Fresnel reflectance, and is the de facto standard in physically based rendering. Ref-NeRF [11]

reparametrizes outgoing radiance as reflected diffuse and specular components, showing cleaner geometry when the radiance field respects reflection geometry. These methods either tie material modeling to a radiance field rather than an explicit surface or assume per-scene optimization with no path to feed-forward inference on unseen objects.

3. Methodology

This section investigates whether photometric supervision through a differentiable physics-based renderer can identify Cook-Torrance BRDF parameters and a parametric light model jointly with an SDF surface, given sparse multi-view RGB plus per-frame depth and normals. It is organized around four design questions: differentiable implicit-surface rendering, view conditioning without per-scene optimization, identifiable material and lighting parameterization, and balancing OmniObject3D's supervision signals.

3.1. Architecture

Image encoder

A ResNet-34 backbone pretrained on ImageNet processes each input view, splitting into two paths. A shallow path through the stem and first two residual groups produces $H/8 \times W/8$ feature maps, projected to 16 channels for PixelNeRF-style local projection. A deep path through layer3, layer4, and adaptive average pooling yields a 512-d per-view descriptor, projected via LayerNorm-linear to a global latent $z \in \mathbb{R}^{128}$ that conditions the BRDF and light heads. Per-view descriptors are mean-pooled when $N > 1$. The encoder is fine-tuned at lr $1e-4$ to preserve ImageNet features.

Neural SDF MLP

The SDF MLP maps a 3D query x , global latent z , and optional locally projected features to a scalar signed distance $f(x)$. Positional encoding uses 5 Fourier frequencies (33-d); local features are concatenated at the input. Rather than concatenating the global z at every layer (which weakens its influence), z conditions each hidden layer through FiLM modulation, with γ , β initialized to (1, 0). The MLP has 8 layers with a skip at layer 4, hidden dim 256, and sphere-initialized weights $f(x) \approx \|x\| - 0.35$.

Local feature projection follows PixelNeRF [13]: each 3D query is projected into every input view, the shallow feature map is bilinearly sampled, and features are averaged across views before reaching the SDF MLP — supplementing global z conditioning with spatially localized context.

BRDF and light heads

Two MLP heads (two layers, width 128, ReLU) decode z . The BRDF is Cook-Torrance GGX (albedo, roughness, metalness) [2] — the standard physical material model — which handles diffuse and specular regimes and degenerates to nearly Lambertian when roughness saturates. The BRDF head outputs five sigmoid values; metalness is initialized near 0.1 reflecting the dielectric prevalence in OmniObject3D. The light head outputs a 3D point-light position, 3D intensity (softplus, clamped [0, 20]), and 3D ambient fill (softplus, clamped [0, 10]). Final linear weights zero-initialize so initial predictions depend only on biases.

BRDF and light parameters are spatially uniform per object: identifying spatially varying reflectance from sparse views is ambiguous without a strong prior (a large GT-BRDF dataset or pretrained material decoder), and a single (albedo, roughness, metalness) triple still captures the specular-vs-diffuse axis distinguishing broad material classes — the principal failure mode of Lambertian reconstruction. Spatially varying reflectance is the most impactful extension and is discussed in Section 6.1.

Differentiable NeuS rendering

For each ray, PRISM uses stratified sampling (96 in [0.5, 6.0]) plus hierarchical importance resampling (24 extra). NeuS [8] is chosen over IDR sphere tracing because it produces non-zero gradients through silhouettes — essential when the SDF is coarsely initialized — and samples color and depth at every point along the ray, coupling cleanly with PixelNeRF-style local feature projection. Weights use the logistic-density approximation with learned sharpness $\beta = \exp(\log_beta)$.

β concentrates rendering weights around the surface. I anneal its upper bound from 0.25 to $1e-6$ over 30 epochs: a soft surface early in training prevents collapse into degenerate minima, sharpening once geometry is approximately right.

Expected depth is $d = (\sum w_i t_i) / (\sum w_i)$, with rays below $\sum w_i = 0.15$ treated as empty. Expected normals

are $\text{normalize}(\sum w_i \nabla f(x_i))$ from autograd. Cook-Torrance GGX shading is evaluated at $x_s = \text{origin} + d \cdot \text{direction}$ using the predicted BRDF and light.

3.2. Training Objective

The total loss sums sixteen terms in five groups: photometric, geometric, SDF regularity, free-space and silhouette, and one optimization helper. Loss weights λ are in config.py.

Photometric. L_{render} ($\lambda = 0.35$) is L1 between predicted Cook-Torrance shaded color and ground-truth RGB at foreground pixels. $L_{\text{perceptual}}$ ($\lambda = 0.5$) computes L1 in the relu1_2 and relu2_2 layers of a pretrained VGG-16 on a 32×32 patch, encouraging mid-level structure like edges and textures rather than raw pixel matches.

Geometric. L_{depth} ($\lambda = 4.0$) applies L1 between rendered and GT depth where GT is valid; L_{normal} ($\lambda = 2.5$) applies $1 - \cos$ similarity between rendered and GT normals. These dominate λ because OmniObject3D supplies dense depth and normal ground truth, and learning shape directly is far more sample-efficient than learning it indirectly from photometric supervision. The render loss is weighted lower because its role is to identify BRDF and lighting parameters, which have no direct supervision.

SDF regularity. L_{eikonal} ($\lambda = 1.0$) enforces $|\nabla f| = 1$ so the network learns a true SDF — without this, volume rendering weights become unreliable. Three additional terms ground the geometry at complementary scales: $L_{\text{sdf_surface}}$ pins the zero level set to the GT surface, $L_{\text{sdf_sign}}$ enforces correct sign along the full ray, and $L_{\text{sdf_band}}$ applies the sign constraint in a narrow band around the surface.

Free-space and silhouette. Background rays carry $L_{\text{bg_sdf}}$ (push SDF positive) and $L_{\text{bg_alpha}}$ (penalize accumulated weight). A combined BCE+Dice silhouette loss acts on a soft silhouette derived from the minimum SDF along each ray. L_{hull} implements multi-view space carving — any 3D point projecting outside the foreground mask in any input view is penalized — with masks from flood-fill (sufficient for OmniObject3D's clean backgrounds). A weak closure prior keeps the SDF negative at the object center and positive near the reconstruction boundary, preventing degenerate open surfaces.

Optimization helper. $L_{\text{light_facing}}$ ($\lambda = 0.5$) penalizes normals whose dot product with the predicted light direction is negative, preventing a degenerate configuration where normals point away from the light, producing zero specular gradients and stalling BRDF learning.

3.3. Training Configuration

PRISM is trained for up to 100 epochs at batch size 4 and 256×256 input resolution. AdamW uses $\text{lr} = 5e-4$ ($1e-4$ for the encoder), weight decay $1e-4$, cosine schedule with 500 warmup steps. Each step samples 512 rays per image, plus 1,024 patch pixels for the perceptual loss on one batch element. Gradient clipping at norm 1.0 stabilizes training. AMP is used for the encoder and BRDF/light heads; the SDF MLP and gradient computations remain in float32 to prevent overflow in the positional encoding and eikonal terms.

The 100-epoch ceiling is a compute-budget choice rather than a convergence point: the total training loss was still trending downward when training was cut off. Reaching wall-clock budget at this cap, combined with the need to evaluate ablation variants under matched conditions, was the binding constraint that forced the three architectural ablations described in Section 5.5 to be deferred. With more compute, training longer and at higher resolution with more rays and more samples per ray would let the model resolve fine surface detail from the rendered views — the dominant remaining headroom on appearance after spatially varying reflectance.

3.4. Evaluation Protocol

Geometry is evaluated by marching cubes at 192^3 in $[-2.5, 2.5]^3$, followed by 5 passes of Laplacian smoothing and largest-connected-component selection. Meshes are normalized to GT bounding-box scale before metrics. Chamfer is the mean bidirectional nearest-neighbor distance between 100k-point uniform samples. PSNR uses foreground pixels of rendered novel views (background excluded to prevent trivial inflation from black-background agreement). Aggregate metrics are means and standard deviations across objects.

4. Results and Discussion

4.1. Experimental Setup

All experiments use OmniObject3D [1]: blender_render_24_views supplies 24 RGB images per object at 800×800 with per-frame depth, normals, extrinsics, and masks; raw_scans supplies textured OBJ meshes as geometric ground truth. PRISM uses $N = 5$ input views; the remaining 19 views per object are held out for novel-view PSNR. The full model is evaluated on 19 held-out objects across 6 categories (10 boxes, 3 bottles, 2 clocks, 2 cups, 1 chess piece, 1 dumbbell). Loss-group ablations use a 5-object subset under matched configurations. All metrics come from a single trained model per condition; multi-seed variance is left to future work.

4.2. Quantitative Results

Table 1 reports the aggregate metrics for the full PRISM model across the 19-object test set, and Table 2 reports per-category breakdowns for the four categories with at least two test objects.

Table 1: Aggregate metrics for the full PRISM model on the 19-object held-out test set. Chamfer is reported in normalized units (meshes rescaled to GT bounding box); PSNR is computed on foreground pixels of rendered novel views.

Metric	Mean	Std
Chamfer distance ↓	0.209	0.043
Foreground PSNR (dB) ↑	14.55	2.52

Table 2: Per-category breakdown across categories with $n \geq 2$ test objects. PSNR varies more across categories than Chamfer does, consistent with the spatially uniform BRDF being the dominant source of appearance error.

Category	n	Chamfer ↓	PSNR (dB) ↑
box	10	0.219 ± 0.046	13.72 ± 1.96
bottle	3	0.227 ± 0.013	16.45 ± 2.68
clock	2	0.167 ± 0.016	13.12 ± 0.40
cup	2	0.186 ± 0.054	14.67 ± 1.40

Two patterns stand out. First, Chamfer is tightly clustered (std 0.043 over a mean of 0.209, ~21% relative spread), and per-category Chamfer means are similarly close — geometric supervision drives the SDF to broadly comparable accuracy across diverse shapes. Second, PSNR varies substantially more: the 2.52 dB std and 9.3

dB gap between worst (box_079, 11.06 dB) and best (dumbbell_002, 20.40 dB) is much larger than what a primarily geometry-driven appearance model would predict. The highest-PSNR objects have the most uniform surface color (dumbbell, bottle); the lowest-PSNR have the most varied texture (multi-color boxes). This is the signature of the global BRDF limitation in Section 6.1: geometry generalizes across categories, but a single-color appearance model cannot.

4.3. Qualitative Results

Figure 1 shows the five input views supplied to PRISM for clock_029, and Figure 2 shows the extracted mesh after marching cubes. Numerically, clock_029 is one of the better geometric reconstructions in the test set (Chamfer 0.150, well below the 0.209 mean) while sitting near the lower end on appearance (PSNR 12.72 dB).



Figure 1: The five input views (v0-v4) provided to PRISM for clock_029. The model receives only these five RGB images plus their camera parameters; novel-view PSNR is then evaluated on the remaining 19 views per object.

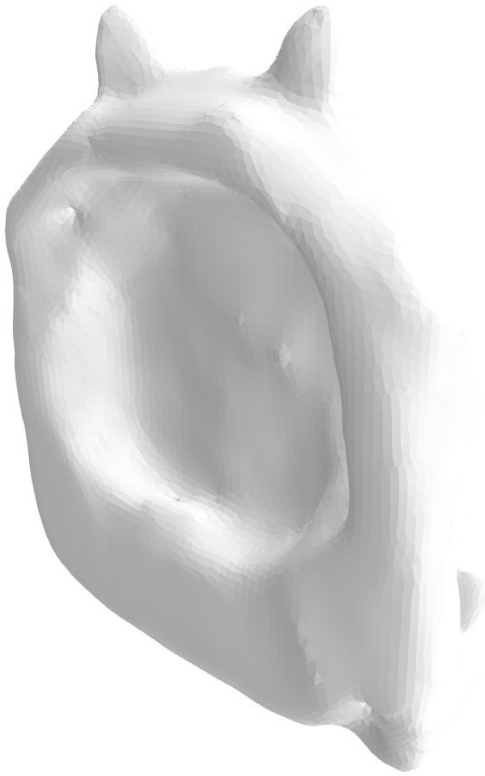


Figure 2: Extracted mesh for clock_029 (marching cubes at 192^3 , 5 passes Laplacian smoothing, largest-connected-component). The overall silhouette and dominant geometric features are recovered, including the recessed face impossible to capture through silhouette analysis alone.

The mesh recovers the dominant structure of the clock — rounded body, two ear-like protrusions, and the recessed face — but fine surface relief (numerals, button) is smoothed over. This is expected given the 192^3 marching-cubes resolution and the fact that only the photometric loss could pick up sub-millimeter detail, while depth and normal losses (which dominate λ) encode bulk geometry. A slight body compression relative to v0 is also visible.

Per-view PSNR for clock_029 ([12.4, 15.6, 13.2, 11.3, 11.1] dB across v0–v4) shows substantial view-to-view variance, with v1 — the most face-on dial view — scoring highest. This is the global-BRDF effect in microcosm: the model picks the single albedo that best explains the clock’s average color, so views dominated by the off-white body score higher than views showing more of the secondary-color top.

5. Ablation Studies

To quantify the contribution of each loss group, I trained four ablation variants on a 5-object subset under otherwise identical conditions. Each variant zeros the λ for one loss group; all other hyperparameters and the input view set are held fixed. The 5-object baseline controls for the difference between this subset and the full 19-object evaluation.

Table 3: Loss-group ablations on a fixed 5-object subset. PSNR drop is computed against the 5-object baseline (13.17 dB), not the 19-object full-model number. Chamfer is essentially constant across these conditions, indicating that on this subset the variations in geometric supervision do not change overall geometric accuracy at the resolution measured by Chamfer.

Condition	Chamfer ↓	PSNR (dB) ↑	Δ PSNR vs baseline
Full model (19 obj)	0.209 ± 0.043	14.55 ± 2.52	—
Baseline (5 obj subset)	0.219 ± 0.054	13.17 ± 0.73	—
No photometric loss	0.219 ± 0.054	8.69 ± 1.94	−4.48
No depth supervision	0.219 ± 0.054	12.03 ± 0.41	−1.14
No normal supervision	0.219 ± 0.054	12.94 ± 0.76	−0.23
No eikonal regularizer	0.219 ± 0.054	13.09 ± 1.07	−0.08

5.1. Photometric supervision is dominant for appearance

Removing L_{render} and $L_{\text{perceptual}}$ drops PSNR by 4.48 dB, by far the largest effect. Without photometric gradients, the BRDF and light heads drift toward initialization. PSNR does not collapse entirely (8.69 dB) because the geometric losses still produce a roughly correct surface and the BRDF/light biases give a default gray-ish shading that partially matches OmniObject3D content.

5.2. Depth supervision is the next most important contributor

Removing L_{depth} drops PSNR by 1.14 dB — entirely indirectly, through the geometry: a worse SDF shifts the shading sample $x_s = \text{origin} + d \cdot \text{direction}$ and degrades Cook-Torrance evaluation.

5.3. Normal supervision and eikonal regularization have small effects

Removing L_{normal} drops PSNR by 0.23 dB and removing L_{eikonal} by 0.08 dB. The small normal effect is consistent with depth already constraining the SDF gradient direction implicitly. The eikonal result is more interesting: the architecture works almost as well without explicit $|\nabla f| = 1$, suggesting depth and sign-band losses produce SDF values well-behaved enough for NeuS rendering. I retain eikonal for its negligible cost and theoretical grounding, but it is not load-bearing on this dataset.

5.4. Chamfer is constant across loss-group ablations

Chamfer is identical to within reported precision (0.219 ± 0.054) across all four ablation variants on the same 5-object subset. On these objects bulk geometry is determined primarily by the silhouette, free-space, and SDF sign losses — none of which is removed here — so the four removed terms each contribute to finer surface detail not captured at the Chamfer measurement scale. A higher-resolution geometric metric would be needed to distinguish them.

5.5. Architectural ablations not reported

Three architectural ablations were planned — single-view ($N=1$) vs. multi-view, global-only vs. local-feature conditioning, and fixed vs. learned light — but were not run. The loss was still trending downward at the 100-epoch cap (Section 3.3), and the wall-clock budget needed to train each of the four loss-group variants to that cap left no compute headroom to also train the three architectural variants. They are deferred to future work; the loss-group results above already establish a clear pecking order — photometric $\gg$ depth $\gg$ normal $\approx$ eikonal $\approx$ 0 for appearance.

6. Failure Cases and Limitations

6.1. Spatially Uniform BRDF

The most significant architectural limitation is the global, spatially uniform material model. Virtually every real object has non-uniform surface color, yet PRISM predicts a single albedo per object. The best the model can do is match the average color, which inflates photometric

loss wherever surface color deviates from that average. Roughness and metalness are affected too, but the albedo gap dominates perceptual error. The test results confirm this: the highest-PSNR objects (dumbbell, bottle) have the most uniform color, the lowest-PSNR (multi-color box packaging) the most varied.

The focus of this work is geometry and surface normals rather than appearance. A spatially varying albedo would require a texture field or neural appearance model, shifting the problem toward view synthesis. Per-point albedo prediction is the single most impactful future extension.

6.2. Limited Compute

The second-most impactful extension is more compute. The total loss was still falling at the 100-epoch cap (Section 3.3), so longer training is expected to yield further gains. With a larger budget, raising input resolution above 256×256 and increasing rays and samples per ray would let the photometric path resolve sub-millimeter detail that depth and normal supervision alone cannot reach — the smoothing of the clock numerals and button in Figure 2 is the principal symptom of this ceiling. These extensions address the dominant residual error sources from the compute ceiling.

6.3. Point Light Approximation

A single point light plus isotropic ambient fill cannot represent OmniObject3D's Blender HDRI environment lighting, which is spatially varying and multi-directional. The light parameters can only converge to a plausible approximation, leaving residual ambiguity in BRDF estimates where the true illumination differs significantly from a point source. Replacing the LightHead with a learned environment representation [12] is a natural future direction.

6.4. Thin Structures

NeuS rendering with 120 total ray samples can miss thin structures whose SDF zero crossing falls in a narrow gap between samples. Hierarchical importance resampling helps but does not eliminate the artifact; raising n_{samples} reduces it at quadratic memory cost.

6.5. Loss Weight Sensitivity

Sixteen loss terms with independent λ values create a complex optimization landscape. Conflicts — for instance

render and `sdf_sign` disagreeing when depth is slightly mis-registered with RGB — can slow convergence. The current λ configuration was tuned manually on validation; gradient-based meta-learning of loss weights is a natural extension.

6.6. Scale and Calibration

PRISM assumes known camera intrinsics and extrinsics, as provided by OmniObject3D. Under approximate or unknown calibration, metric depth predictions become unreliable. In-the-wild use would require a version that estimates or marginalizes over camera parameters.

7. Ethical and Societal Considerations

Physics-informed 3D reconstruction has broad beneficial applications in robotics, accessibility, and medical imaging, but several risks merit consideration. Privacy and surveillance: a system that recovers 3D geometry and material properties from a few images could be used to reconstruct detailed models of individuals or private spaces without consent. PRISM is trained on objects rather than people, but the methodology transfers, and deployment should include safeguards against unauthorized reconstruction of private subjects.

Synthetic media: high-quality reconstruction with plausible materials and relighting is an enabling technology for photo-realistic synthetic imagery and could be repurposed for deceptive content. Responsible deployment should include usage guidelines prohibiting such applications.

Dataset bias: OmniObject3D's 190 categories may underrepresent objects from specific cultural or geographic contexts, yielding systematically lower reconstruction quality for those types. Broader dataset coverage would partially mitigate this.

Safety-critical deployment: reconstruction errors could propagate into downstream decisions in robotics. PRISM provides no uncertainty estimates, so the model cannot signal when its reconstruction is unreliable. Uncertainty quantification (ensemble or variational) should precede any safety-critical deployment.

8. Conclusion

PRISM is a feed-forward multi-view 3D reconstruction system that jointly infers geometry, surface materials, and lighting by coupling a neural SDF with Cook-Torrance GGX BRDF parameters through differentiable NeuS volume rendering. The motivation — that recovering relightable, physically interpretable 3D from sparse views requires more than NeRF-style emitted color or geometry-only SDFs — is addressed by embedding a physically principled BRDF directly in the rendering loop, so material and lighting parameters are identified from pixel observations in the same forward pass that produces the surface.

On OmniObject3D, the full model achieves a mean Chamfer distance of 0.209 and mean foreground PSNR of 14.55 dB across 19 objects in 6 categories. Loss-group ablations reveal a clear ordering: photometric supervision contributes ~ 4.5 dB of PSNR, depth ~ 1.1 dB, and normal and eikonal terms under 0.25 dB each. Qualitatively (clock_029), the architecture recovers bulk geometry while smoothing fine surface relief the global BRDF cannot explain photometrically. The most consistent driver of low PSNR is intra-object color variation that a single-albedo BRDF cannot represent — exactly the spatially uniform material limitation.

Spatially varying albedo is the most direct path to closing the appearance gap, grounded in the empirical signal that PSNR tracks intra-object color variation. Second priority is more compute: the loss had not converged at the 100-epoch cap, and training longer at higher resolution with more rays and samples per ray is the most likely route to recovering fine surface detail. Richer environment lighting, uncertainty quantification, the deferred architectural ablations, and operation under unknown camera calibration follow. Together these define a concrete path from a working physics-informed reconstruction system on curated benchmarks toward one that can produce relightable, physically interpretable 3D from sparse images in the wild — the original motivation of the work.

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AI Use Statement

Google Gemini was used to help parse technical references — given the 30-hour target scope of the project, the LLM was useful in providing a competent understanding of the prior computer vision and graphics works. Cursor was used to help orchestrate the multi-file codebase and manage my developing fluency in PyTorch.